

Statistics refresher

International Marketing Analytics



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Scale types

Centre and
spread

Normal
distribution

Covariance
and
correlation

Python in
Positron

The concepts



- Scale types: what you may calculate
- Mean, median, variance, standard deviation
- Standard error and standardisation (z)
- Normal and standard normal distribution
- Covariance and correlation

The practice



- One small example you can follow with a calculator
- Every formula by hand on the slide
- The same steps by hand in Python
- Check against polars' built-in functions



Why now: regression output is made of these pieces: means, standard errors, z-values and correlations.

Our example: a chocolate kiosk

Five weeks, five prices

Week	Price (EUR)	Bars sold
1	2.00	110
2	2.50	100
3	3.00	80
4	3.50	70
5	4.00	40

One week, day by day

Day	Bars sold
Mon	10
Tue	12
Wed	11
Thu	13
Fri (event)	54



Small on purpose: every number today can be checked with a calculator, then in Python.

Scale types



Part 1 of 5 · nominal, ordinal, interval, ratio

Four scale types

Scale	What you can say	Kiosk example	Typical statistics
Nominal	same or different	brand, payment method	counts, mode
Ordinal	more or less	rating 1 to 5 stars, size S/M/L	median, percentiles
Interval	how much more (no true zero)	temperature in °C	mean, standard deviation
Ratio	how many times (true zero)	price, bars sold, revenue	all, incl. ratios



Two groups: nominal and ordinal are categorical; interval and ratio are metric (numbers you can average).

In regression: categorical variables enter as dummies (0/1), metric variables as they are.

Why the scale matters

20 °C is not twice as warm as 10 °C: no true zero

Mean of star ratings: treats ordinal as metric; say so

Store ID 17: a number, but only a label: nominal

Quick quiz: which scale?

- Country of the customer
- Satisfaction from 1 to 7
- Age in years
- Discount: yes or no
- Temperature on the day



Rule: check the scale before you average, chart or put a variable into a model.

Centre and spread



Part 2 of 5 · mean, median, variance, standard deviation, standard error

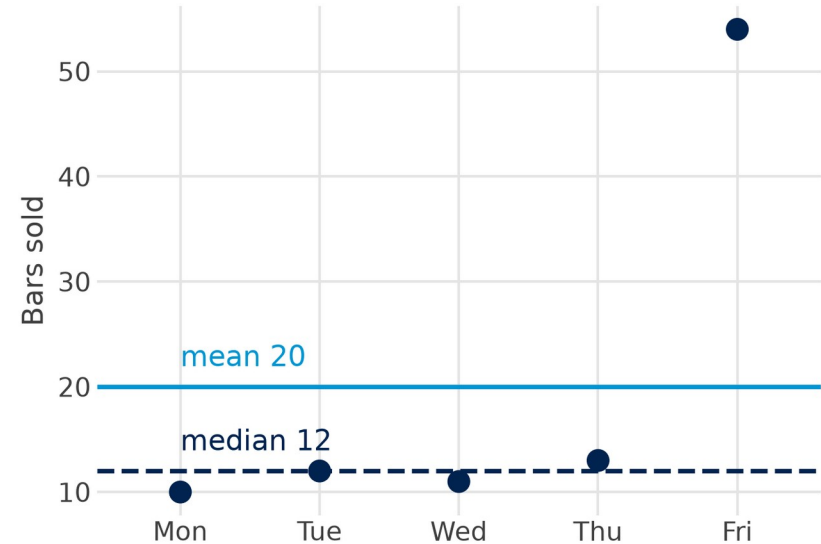
Mean and median

mean $\bar{x} = \Sigma x / n = 100 / 5 = 20$
median middle of 10 11 12 13 54 = 12

Mean uses every value; one event day pulls it up

Median is robust: half the days below, half above

Use the median for skewed data and ordinal scales



Totals need the mean: 20 bars a day $\times$ 5 days = 100 bars in the week.

Variance and standard deviation, by hand

Week	Bars x	$x - \bar{x}$	$(x - \bar{x})^2$
1	110	30	900
2	100	20	400
3	80	0	0
4	70	-10	100
5	40	-40	1,600
Sum	400	0	3,000

Variance and standard deviation

$$\begin{aligned}\bar{x} &= 400 / 5 = 80 \\ s^2 &= 3,000 / (5 - 1) = 750 \\ s &= \sqrt{750} = 27.4 \text{ bars}\end{aligned}$$

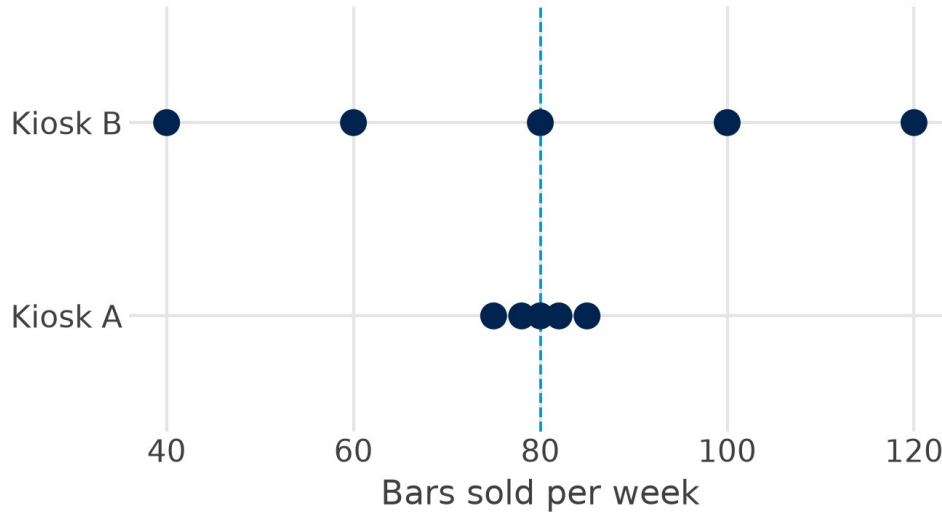
Variance is in squared units (bars²)

Standard deviation is in bars: the typical distance from the mean



Why $n - 1$? deviations from the sample mean are a little too small; dividing by $n - 1$ corrects this. polars and pandas do it by default.

Same mean, different spread



3.8

bars: sd of kiosk A

31.6

bars: sd of kiosk B



For the manager: both sell 80 a week, but kiosk B needs much more safety stock and its forecasts are less certain.

Standard error: how precise is the mean?

$$SE = s / \sqrt{n}$$

$$95\% \text{ confidence interval} \approx \bar{x} \pm 1.96 \cdot SE$$

EUR 1.00

SE with 16 customers: $4 / \sqrt{16}$

EUR 0.50

SE with 64 customers: $4 / \sqrt{64}$

10.04-13.96

95 % interval for a mean
spend of EUR 12 ($n = 16$)

sd describes how much customers differ: spending varies by about EUR 4

SE describes how precise the average is: it shrinks with $\sqrt{n}$; four times the sample, half the SE



In every regression table: the column 'std err' is the standard error of a coefficient.

Standardisation and the normal distribution



Part 3 of 5 · z-scores, normal and standard normal distribution

Standardisation: z-scores

$z = (x - \bar{x}) / s$ a standardised variable has mean 0 and standard deviation 1

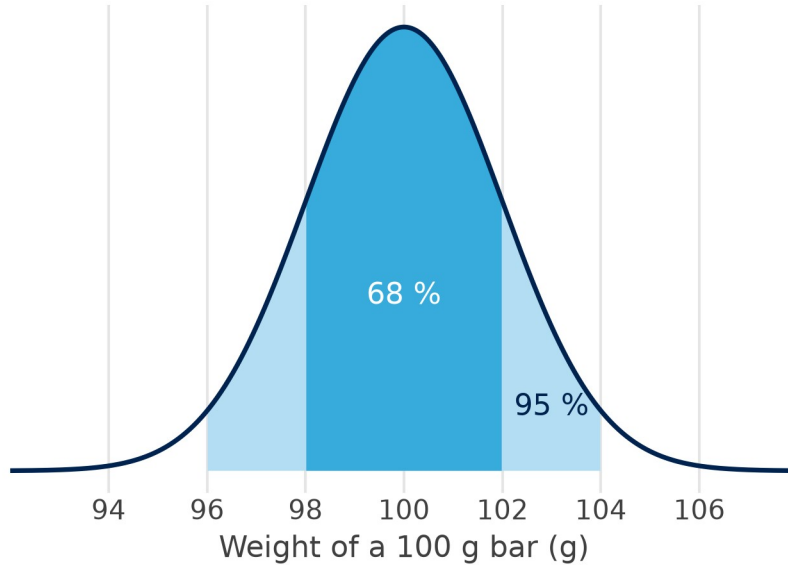
Kiosk	Bars this week	Mean	sd	z
Vienna	140	120	10	+2.0
Graz	75	60	15	+1.0
Our kiosk, week 5	40	80	27.4	-1.46



Read it: Vienna's week lies 2 standard deviations above normal, Graz's only 1: Vienna's week was the more unusual one.

Use: compare across stores, countries or units; spot outliers ($|z| > 3$); standardised coefficients.

The normal distribution



68 %

within ± 1 sd: 98 to 102 g

95 %

within ± 2 sd: 96 to 104 g

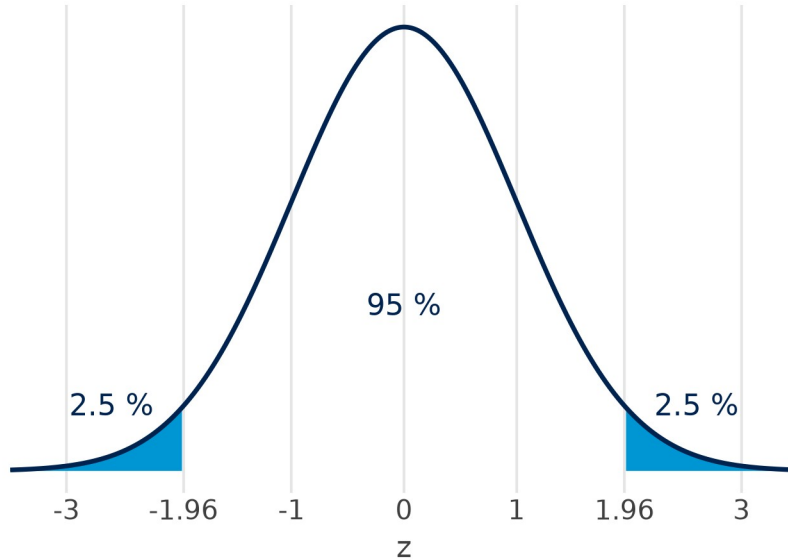
99.7 %

within ± 3 sd: 94 to 106 g



Why it matters: averages and regression coefficients are approximately normal (central limit theorem), so we can test them.

The standard normal distribution



z	share below z
1.00	84.1 %
1.645	95.0 %
1.96	97.5 %
2.576	99.5 %

Bars under 97 g? $z = (97 - 100) / 2 = -1.5$,
so 6.7 % of all bars



1.96: the number behind 95 % confidence intervals and two-sided tests at 5 %.

Covariance and correlation



Part 4 of 5 · do two variables move together,
and how strongly?

Covariance: do price and sales move together?



Week	$p - \bar{p}$	$x - \bar{x}$	product
1	-1.0	30	-30
2	-0.5	20	-10
3	0	0	0
4	0.5	-10	-5
5	1.0	-40	-40
Sum			-85

$$\text{cov} = -85 / (5 - 1) = -21.25$$

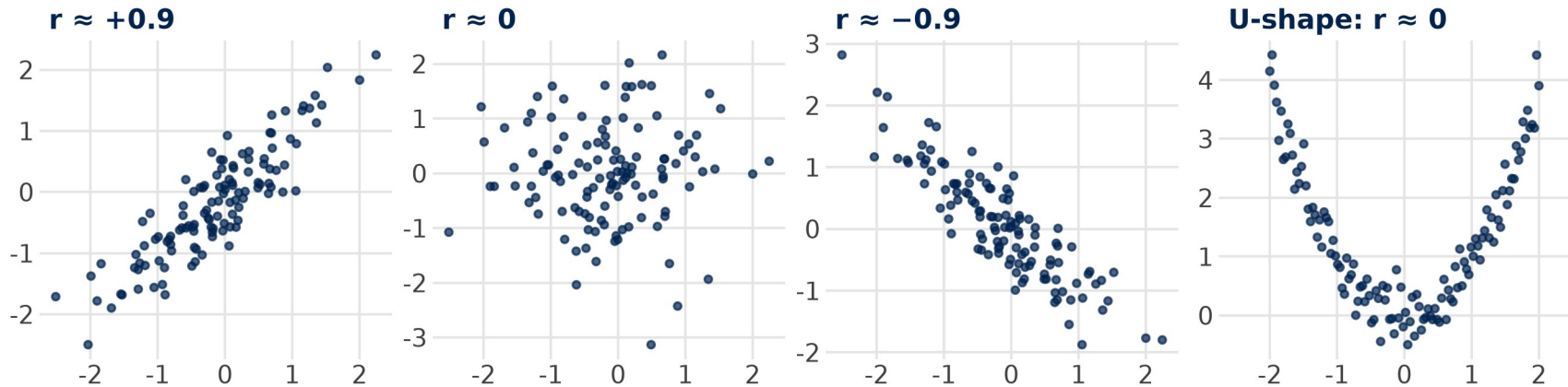


Sign = direction; the size depends on the units (EUR × bars).

Correlation: covariance without units

$$r = \text{cov} / (s_p \cdot s_x) = -21.25 / (0.79 \cdot 27.39) = -0.98$$

$$-1 \leq r \leq +1$$



Two warnings: r only measures straight-line relationships (the U has $r \approx 0$), and correlation is not causation.

Sources: Bojinov, Parzen & Hamilton 2025 (HBS note 9-622-100)

From correlation to regression

Not causation: ice-cream sales and sunburn rise together; sunshine drives both

Correlation says how strongly two variables move together, from -1 to $+1$

Regression says by how much y changes when x changes by one unit

slope $b = \text{cov}(p, x) / \text{var}(p) = -21.25 / 0.625 = -34$ bars per EUR
line through the means: $x = 80 - 34 \cdot (p - 3)$



Next: the least-squares line, built from exactly these pieces.

Python in Positron



Part 5 of 5 · mean, standard deviation, z-scores, covariance and correlation by hand, then checked

Step 1: the data and the mean

Python: polars

```
import polars as pl

kiosk = pl.DataFrame({
    "week": [1, 2, 3, 4, 5],
    "price": [2.0, 2.5, 3.0, 3.5, 4.0],
    "units": [110, 100, 80, 70, 40],
})

n = kiosk.height                # 5
mean_units = kiosk["units"].sum() / n  # by hand
kiosk["units"].mean()            # check
kiosk["units"].median()
```

Result	Value
n	5
mean, by hand	80.0
mean(), check	80.0
median()	80.0



Run it: click into a cell and press Ctrl/Cmd + Enter; the result appears in the Console.

Step 2: variance, standard deviation, standard error

Python: by hand, then checked

```
dev = kiosk["units"] - mean_units
dev          # 30, 20, 0, -10, -40

var_units = (dev ** 2).sum() / (n - 1)    # 750.0
sd_units = var_units ** 0.5              # 27.39
kiosk["units"].std()                     # check: n - 1

se_units = sd_units / n ** 0.5           # 12.25
```

Result	Value
variance	750.0
sd, by hand	27.39
std(), check	27.39
standard error	12.25



Note: std() divides by $n - 1$ (ddof=1), like the formula on the slide.

Step 3: standardise

Python: z-scores

```
kiosk = kiosk.with_columns(  
    z_units=(pl.col("units") - pl.col("units").mean())  
            / pl.col("units").std()  
)  
kiosk  
  
kiosk["z_units"].mean()    # 0  
kiosk["z_units"].std()     # 1
```

Week	Bars	z
1	110	1.10
2	100	0.73
3	80	0.00
4	70	-0.37
5	40	-1.46



Read it: week 5 lies 1.46 standard deviations below an average week.

Step 4: covariance and correlation

Python: by hand, then checked

```
dev_p = kiosk["price"] - kiosk["price"].mean()
dev_u = kiosk["units"] - kiosk["units"].mean()

cov = (dev_p * dev_u).sum() / (n - 1)          # -21.25
r = cov / (kiosk["price"].std()               # -0.981
        * kiosk["units"].std())

kiosk.select(
    pl.cov("price", "units").alias("cov"),
    pl.corr("price", "units").alias("r"))      # check
```

Result	Value
covariance	-21.25
correlation	-0.981
pl.cov, check	-21.25
pl.corr, check	-0.981



Same numbers as by hand: the computer only saves the typing.

Step 5: look at it

Python: plotnine

```
from plotnine import (ggplot, aes,  
                      geom_point, labs)  
  
(ggplot(kiosk, aes("price", "units"))  
 + geom_point(size=3)  
 + labs(x="Price (EUR)",  
        y="Bars sold per week"))
```



Plot first: a correlation can hide a curve, an outlier or two groups.

Your turn: by hand, then in Python

**Pairs, paper:
10 minutes**

**Python check:
10 minutes**

Day	Temp. °C	Bars
Mon	10	60
Tue	15	55
Wed	20	50
Thu	25	40
Fri	30	35

- 1 Scales:** which scale is each column?
- 2 Centre:** mean and median of bars sold
- 3 Spread:** variance and sd of bars, by hand
- 4 z:** z-score of the hottest day's sales
- 5 Together:** covariance and r of temperature and bars; then check all in Python

Key takeaways

- 1 Scale** first: it decides which statistics make sense
- 2 Mean and median:** the median resists outliers; totals need the mean
- 3 sd** describes the data; $SE = sd / \sqrt{n}$ describes the precision of the mean
- 4 $z = (x - \bar{x}) / s$** makes values comparable; 95 % of a normal lies within ± 1.96 sd
- 5 Covariance** gives the direction, correlation the strength (-1 to $+1$), regression the size



Next: linear regression on real chocolate data.

Books and readings



Field, A. (2024). Discovering Statistics Using IBM SPSS Statistics, 6th ed. SAGE. Chapters 1 to 2 and 8.

Bojinov, I. I., Parzen, M. & Hamilton, P. J. (2025). Linear Regression. HBS note 9-622-100: correlation.

Hair, J. F. et al. (2019). Multivariate Data Analysis, 8th ed. Cengage. Chapter 2.

Stock, J. H. & Watson, M. W. (2019). Introduction to Econometrics, 4th ed. Pearson. Chapters 2 to 3.

Online



Seeing Theory. Brown University: interactive visual introduction to probability and statistics (free).

StatQuest. Videos on the normal distribution, standard deviation vs standard error, covariance and correlation (free).

polars documentation. mean, median, std, cov, corr.

Gelman, Hill & Vehtari (2020). Regression and Other Stories, ch. 3 to 4 (free PDF).



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